Executive Report: SurplusSense Decision Support

SMU MBA Machine Learning — Individual Project | April 2026

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1. Executive Summary

SurplusSense helps food merchants convert end-of-day surplus uncertainty into faster, safer, and more margin-aware decisions. It is a decision-support product — not a dashboard — that translates surplus predictions into specific recommended actions under food-safety guardrails. The product is pilot-ready; commercial deployment requires a structured merchant pilot with 3–5 outlets.

Singapore generated 784,000 tonnes of food waste in 2024 (National Environment Agency). The merchant-side surplus decision — hold, monitor, discount, deep discount, donate, or discard — is where value is lost or recovered. Existing platforms help consumers discover discounted surplus; none help merchants decide what to discount, at what intensity, or when. SurplusSense fills this gap.

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2. Problem Worth Solving

For F&B merchants — bakeries, cafés, and prepared-food outlets — the problem is a daily operational decision at 4pm: **what do I do with unsold inventory?**

The decision is genuinely hard:

- **Time pressure**: shelf life is already running down
- **Margin trade-off**: deeper discounts recover more revenue but erode margin on items that would have sold at full price
- **Safety risk**: items past safe holding times cannot be listed
- **Labour constraints**: staff must decide under operating pressure
- **Inconsistent judgment**: the same surplus item may receive different decisions across shifts

The monetisable gap is narrower than "food waste" broadly: small F&B merchants often know they may have surplus, but lack a structured way to decide what action to take before the item loses value. The key decision happens late in the operating day. This is not only a forecasting problem — it is a time-sensitive trade-off between revenue recovery, food safety, staff workload, brand risk, and customer experience.

Target buyer: Singapore-based bakery, café, or prepared-food outlet with recurring end-of-day surplus and enough daily volume for recovered value to exceed SGD 99–299 monthly subscription.

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3. Product Approach

SurplusSense is a **decision-support product** — it translates surplus-risk predictions into specific merchant actions. The six-step workflow:

Step	What happens
1 — Enter context	Merchant selects product category, storage type, holding time, shelf life

Step	What happens
2 — Predict surplus	XGBoost model estimates expected surplus units from 31 raw features (expanded to 47 model input columns after one-hot encoding)
3 — Recommend action	Rule engine maps surplus quantity and shelf life to HOLD / MONITOR / DISCOUNT / DEEP DISCOUNT / DONATE / DISCARD
4 — Screen for safety	Five deterministic checks return SAFE / CAUTION / BLOCKED
5 — Estimate recovery	System calculates discounted revenue recovery vs potential loss
6 — Explain recommendation	Merchant sees the specific reason for the recommended action

Design choice: rules over learning. Recommendations are rule-based because food safety cannot be entrusted to an opaque learned policy. A merchant who lists an unsafe item creates liability that no accuracy metric offsets. Rule-based logic is fully explainable — every recommendation traces to a specific rule.

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4. ML/AI Architecture and Results

Why This Is an ML Decision-Support Product, Not a Dashboard

Most F&B tech products show prediction scores. This product shows decisions. The distinction matters:

- A merchant with 11 surplus units of Ciabatta needs to know: **50% discount, list now, SGD 35 recovered** — not "MAE 0.64"
- The prediction layer (XGBoost) estimates *what will happen*; the downstream layers decide *what to do about it*
- Cold-start merchants (no historical data) receive category-benchmark estimates, so the product is useful from day one

The architecture deliberately separates prediction from prescription: **the model estimates what may happen; the rule and safety layers decide what action is permissible and commercially sensible.**

Architecture layers:

Layer	Component	Role
Input	Merchant/item context	Merchant enters product and outlet details
Feature engineering	31 raw features → 47 model input columns	Lag, rolling, expanding-window, temporal, category features
Prediction	XGBoost	Estimates surplus units
Validation	Temporal holdout + TimeSeriesSplit	Forward-looking evaluation
Leakage control	Expanding-window aggregates with shift(1)	Prevents future-information lookahead

Layer	Component	Role
Decision	Rule-based recommendation engine	Converts surplus prediction into action
Governance	Food-safety rules	BLOCK/CAUTION/SAFE overrides commercial optimisation
Output	Merchant decision cockpit	Recommended action + recovery estimate + explanation

Design decisions:

Design decision	Option rejected	Final choice	Why it matters
Train/test split	Random split	Temporal holdout	Prevents inflated accuracy from future information
Feature aggregation	Full-dataset averages	Expanding-window aggregates with shift(1)	Prevents target leakage
Recommendation logic	Learned black-box policy	Transparent rule engine	Safer for food handling; easier for merchants to trust
New merchant handling	Require historical data	Category benchmark fallback	Makes product usable from day one
Output format	Prediction score	Recommended action + recovery value	Matches real merchant decision workflow

Model results (synthetic training data, 4,027 records, 15 merchants, 13 categories, 90-day window):

Model	Temporal Holdout MAE	vs Historical Average
Historical Average	1.49	—
Previous Day	2.01	–35%
XGBoost (Tuned)	**0.64**	**–57%**

5-fold TimeSeriesSplit CV MAE: 1.38 ± 1.22 (higher variance reflects distributional shift in synthetic data).

Validation caveat: Results are from synthetic/prototype data. They indicate technical feasibility and feature-signal logic, not production accuracy. Aggregate features use expanding-window logic to prevent target leakage. Real merchant pilot data is required before commercial deployment.

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5. Business Case

Who pays: Bakeries, cafés, and small F&B operators with 5–50 daily surplus units in Singapore.

Why they pay: Recovered surplus value plus avoided disposal costs could exceed SGD 99–299 per month per outlet, subject to pilot validation.

Merchant unit economics:

Scenario	Daily recovered value	Monthly recovered value (26 operating days)	Supports pricing
Conservative	SGD 6/day	SGD 156/month	Entry plan at SGD 99/month
Base case	SGD 12/day	SGD 312/month	Mid plan at SGD 199/month
Strong case	SGD 20/day	SGD 520/month	SGD 299/month plan + transaction fee

The commercial case does not require every surplus decision to generate large savings. Even SGD 6–12 daily recovery can justify the low-to-mid subscription tier if staff use the system consistently.

Proposed pricing (pilot-stage hypothesis):

Revenue stream	Rationale
Monthly SaaS: SGD 99–299 per outlet	Predictable recurring revenue; easier for merchants to budget
15% recovered-value fee	Aligns SurplusSense revenue with merchant benefit
Future marketplace integration fee	Phase 2 if product connects surplus items to consumer channels
ESG/reporting module	Additional value for chains needing waste reporting across outlets

Buyer-validation table:

Assumption	Current evidence	Validation needed	Success metric
Merchants have surplus pain	Singapore F&B waste 784k tonnes/year (NEA)	4-week pilot with 3–5 merchants	Waste reduction % per outlet
Recommendation acceptance rate	Workflow designed for merchant trust signals	Live pilot observation	Acceptance rate ≥ 60%
Recovered value exceeds cost	Model estimates SGD 6–20/day on prototype	Pilot measures actual recovered revenue	Recovered value ≥ SGD 200/month
Safety gate prevents unsafe listings	5 deterministic rules with BLOCK for high-risk	Rules tested against real operating conditions	Unsafe recommendation rate = 0
Staff use dashboard daily	Low-friction manual entry; no POS required in Phase 1	Pilot observes daily active usage	≥ 4 sessions/week per outlet

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SurplusSense is a pilot-ready decision-support prototype. It is not SFA-validated, and not commercially live. All performance metrics are from synthetic data and require validation before commercial claims.